



Solution microcrystal enables a high-quality CsPbIBr₂ film to minimize intragranular charge recombination for air-stable, carbon-based photovoltaics

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ABSTRACT The colloidal nature of perovskite precursor solution highly determines the quality of perovskite film and the photovoltaic performance of a perovskite solar cell (PSC). Herein, we demonstrate a facile method to fabricate a high-quality all-inorganic CsPbIBr₂ film by regulating the temperature of the precursor solution. After characterization, stable microcrystals with a size up to 1 μm have been formed under high temperatures, more suitable for preferable crystallization by spontaneous nucleation than the precursor solution at room temperature. Consequently, rapid charge transfer across enlarged crystal grain rather than intragranular charge recombination is achieved within the perovskite film because of reduced defects, promoting the efficiency as high as 11.12% for carbon-based PSC, which is higher than that of a control device with 8.51% efficiency. Moreover, the optimal device shows improved stability under air atmosphere and high-temperature conditions.

Keywords: all-inorganic perovskite solar cells, crystal growth, film quality, intermediate phase, stability

INTRODUCTION

In recent years, perovskite solar cells (PSCs) have improved the power conversion efficiency (PCE) from an initial 3.8% to 25.7% [1–3]. To date, their future commercialization is limited by their poor lattice stability under external stimuli [4–6]. In stabilizing the soft perovskite lattice by various strategies, such as compositional engineering and interfacial optimization, all-inorganic perovskites are regarded as potential candidates to balance the incompatible efficiency and stability [7]. Compared with traditional organic-inorganic hybrid perovskites, the large bandgaps of all-inorganic species range from 1.71 eV for CsPbI₃ to 2.30 eV for CsPbBr₃, making them suitable for application in tandem PSCs [8–10]. Among them, halide-mixed CsPbIBr₂ displays robust environmental tolerance [11]. Therefore, increasing efforts to advance the photovoltaic performance of corresponding PSCs is necessary.

Reviewing the highly efficient and stable PSCs, direct-phase conversion from the precursor solution is the mainstream to obtain high-quality perovskite films [12–14]. The physico-

chemical properties of perovskite crystals can be tuned by altering the coordination chemistry in the precursor solution and regulating the crystallization. For example, the solvent with a high coordination ability with the central atom Pb always results in slower film crystallization, whereas the solvent with a low coordination ability delivers a contrary scenario [15]. The mechanism behind this phenomenon is primarily attributed to the formation of colloidal or intermediate complexes [16,17]. Under the following thermal treatment, the supersaturation solution allows the nucleation and crystal growth; however, an energy barrier is always observed for homogeneous nucleation. In addressing this issue, preforming colloids in the precursor solution by using additives or re-dissolving perovskite crystals/powders for hybrid perovskites is a feasible strategy [18,19]. Moreover, comprehensively understanding the effect of precursor properties on the final film quality is important for all-inorganic CsPbIBr₂.

Herein, we demonstrate an effective strategy to preform microcrystals with a size up to 1 μm in the precursor solution by elevating the solution temperature. Large colloids enable the formation of a high-quality CsPbIBr₂ film because of the lowered nucleation barrier. Consequently, a high-quality perovskite film with few defects is obtained. After characterization, the intragranular charge recombination is minimized, leading to an enhanced PCE of 11.12% for carbon-based CsPbIBr₂ PSC, which is the highest value for this kind of PSC. Moreover, the long-term stability of unencapsulated PSC is improved under air atmosphere and high temperatures over 90 days and 400 h, demonstrating the great potential of solution microcrystals in advancing high-performance PSCs.

EXPERIMENTAL SECTION

Materials and reagents

Unless stated otherwise, all materials and reagents were purchased from commercial suppliers and used as received without further purification. Anhydrous dimethyl sulfoxide (DMSO, 99.7%), diethanol amine (DEA, 99%), and lead bromide (PbBr₂, 99.99%) were obtained from Macklin. Titanium (IV) isopropoxide (99.9%) and cesium iodide (CsI, 99.99%) were purchased from Aladdin. Lead iodide (PbI₂, 99.99%) was purchased from

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Xi'an Polymer Light Technology Co., Ltd. Fluorine-doped tin oxide (FTO) glass substrates were obtained from Yinkou Opv-tech Co., Ltd. Carbon paste was purchased from Shanghai MaterWin New Materials Co., Ltd.

Solar cell fabrication

FTO-coated glass with the desired pattern was initially etched with zinc powder and HCl and then thoroughly washed in detergent, acetone, isopropanol, and deionized water. Prior to the fabrication of solar cells, the cleaned FTO was treated with plasma for 180 s, and a compact TiO_2 (c- TiO_2) layer was deposited onto the FTO glass by spin coating an ethanol solution of titanium isopropoxide (0.5 mol L^{-1}) and DEA (0.5 mol L^{-1}) at 7000 r min^{-1} for 30 s and annealed in air at 500°C for 2 h. After treating the FTO/c- TiO_2 substrates *via* oxygen plasma for 3 min, $90 \mu\text{L}$ of 1.0 mol L^{-1} $\text{CsPbI}_2\text{Br}_2$ precursor, in which 367 mg of PbBr_2 and 260 mg of CsI were dissolved in 1 mL of DMSO, was spin-coated onto FTO/c- TiO_2 at 1000 r min^{-1} for 10 s and 3000 r min^{-1} for 50 s at 25, 40, 70, 100, 130, and 160°C , respectively. In particular, the crystallization of perovskite thin film included a two-step thermal treatment. The as-prepared thin film was initially placed on a hotplate at 30°C for 1 min, and then the temperature was increased to 260°C for 5 min to remove the solvent and improve the quality of the perovskite thin film. Finally, a carbon electrode with an average area of 0.09 cm^2 was deposited onto the perovskite film by doctor-blade coating.

Characterization and tests

The morphologies and elemental mapping images of halide perovskites were characterized by an scanning electron microscope (SEM, Hitachi S-4800) equipped with an energy-dispersive spectroscope and transmission electron microscope (TEM, Tecnai G2 F20). The crystal structures were characterized by powder X-ray diffraction (XRD) on a Bruker D8 Advanced diffractometer with a Cu K α radiation source. The photocurrent density-voltage (J - V) curves of solar cells were recorded on an electrochemical workstation (CHI660E) under irradiation of simulated solar light at an optical intensity of 100 mW cm^{-2} (Newport, Oriel Class A, 91195A, AM 1.5G, calibrated by a standard silicon solar cell). The external quantum efficiency (EQE) spectra of various devices were measured by using an IPCE kit from Enli Technology Co., Ltd. equipped with a standard Si crystalline solar cell as a reference. The open-circuit voltage decay curves were measured in an open-circuit mode by illuminating the device under simulated solar light at an optical intensity of 100 mW cm^{-2} for several seconds and then instantaneously switching off the light. The time-resolved photoluminescence (TRPL) decay curves were recorded using a Horiba spectrometer excited by a 500-nm laser, and the PL spectra were recorded *via* a fluorescence spectrometer excited by a 450-nm laser. The Mott-Schottky plots were conducted on an electrochemical workstation (CHI600E, Shanghai Chenhua) at a frequency of 5 kHz with an amplitude of 5 mV in the dark.

RESULTS AND DISCUSSION

Fig. 1a shows the typical protocol for fabricating the inorganic $\text{CsPbI}_2\text{Br}_2$ film by changing the temperature of the precursor solution from 25 to 160°C . The typical Tyndall characteristic shown in Fig. 1b demonstrates that all the solutions contain colloids because of the formation of a light beam with a green

laser to pass through the solution. Compared with the solution at room temperature, the precursor darkens by gradually increasing the temperature to 160°C , which is an indicator of the changed physicochemical features. As shown in Fig. 1c, d, dynamic light scattering (DLS) results demonstrate that the control and high-temperature-treated precursors are colloidal dispersions rather than true solutions with an average size of 1.5 nm, which is consistent with the Tyndall effect. However, the average colloid size is increased to 5 nm, and large colloids are observed in the range of 500–1000 nm in the latter circumstance (130°C). In addition, no large species in the control solution are detected. Based on a previous study [20], iodoplumbate complexes $[\text{PbI}_m\text{X}_n]^{2-m}$ are spontaneously formed in N,N -dimethylformamide (DMF) or DMSO solvent because of the strong Pb-X-Pb corner-sharing coordination interaction. Moreover, high temperatures may play a crucial role in promoting the growth of colloids. Based on the ultraviolet-visible (UV-vis) absorption spectra of all colloidal precursors (Fig. S1), the gradually red-shifted absorption edge around 460 nm with the increase of solution temperature indicates the increased coordination levels [21]. After careful comparison, a slightly increased absorption at the wavelength range of 445–480 nm and reduced absorption at a wavelength of $< 445 \text{ nm}$ are observed, demonstrating that the value of m is preferred to be 4 for high-temperature-treated solution [22,23]. Furthermore, we have characterized the TEM images to comprehensively understand the colloidal structure (the insets of Fig. 1c, d). Consistent with the DLS result, several nanoparticles contained a rod-shaped structure for the precursor solution at room temperature, which can be attributed to the preferable crystallization of PbI_2 with stacked (100) planes along the c -axis direction [21]. For the solution treated at 130°C , some spherical species with a size of $1 \mu\text{m}$ are formed, which can be regarded as the interweaved network of a lead polyiodide framework (Fig. S2). Separating the colloids for TEM characterization is difficult because of the presence of natural crystallization during the sample preparation. We have further characterized TEM mapping images of spherical species to confirm the component of the rod-shaped structure. As shown in Fig. S3, despite the presence of Cs, Pb, Br, and I because of the adsorption of CsBr on the surface of the colloid, the fast Fourier transform image and a plane distance of $3.65 \text{ \AA}$ (Fig. S4) match well with the (100) reflection of the hexagonal PbI_2 phase [24], demonstrating the crucial role of temperature in promoting the formation of solution microcrystals.

In revealing the effect of large microcrystals on perovskite growth dynamics and film quality, we initially monitored the intermediate-phase conversion. As shown in Fig. 2a, both intermediate films from the control and high-temperature-treated precursors show a smooth and uniform surface at the initial stage. However, by prolonging the retention time at room temperature without thermal annealing, no evident change in the control intermediate film was observed within 1 min, whereas the hot solution showed rapid phase conversion at the tenth second. The intrinsic mechanism behind this phenomenon can be attributed to the crystal growth induced by microcrystals because of the reduced energy barrier, which can be cross-checked by the UV-vis absorbance spectra. As shown in Fig. 2b and Fig. S5, the characteristic absorption wavelength of intermediate films is gradually red-shifted with the increase of time, which can be primarily attributed to the changed lattice closer to the cubic $\text{CsPbI}_2\text{Br}_2$ phase. Compared with the control film, the

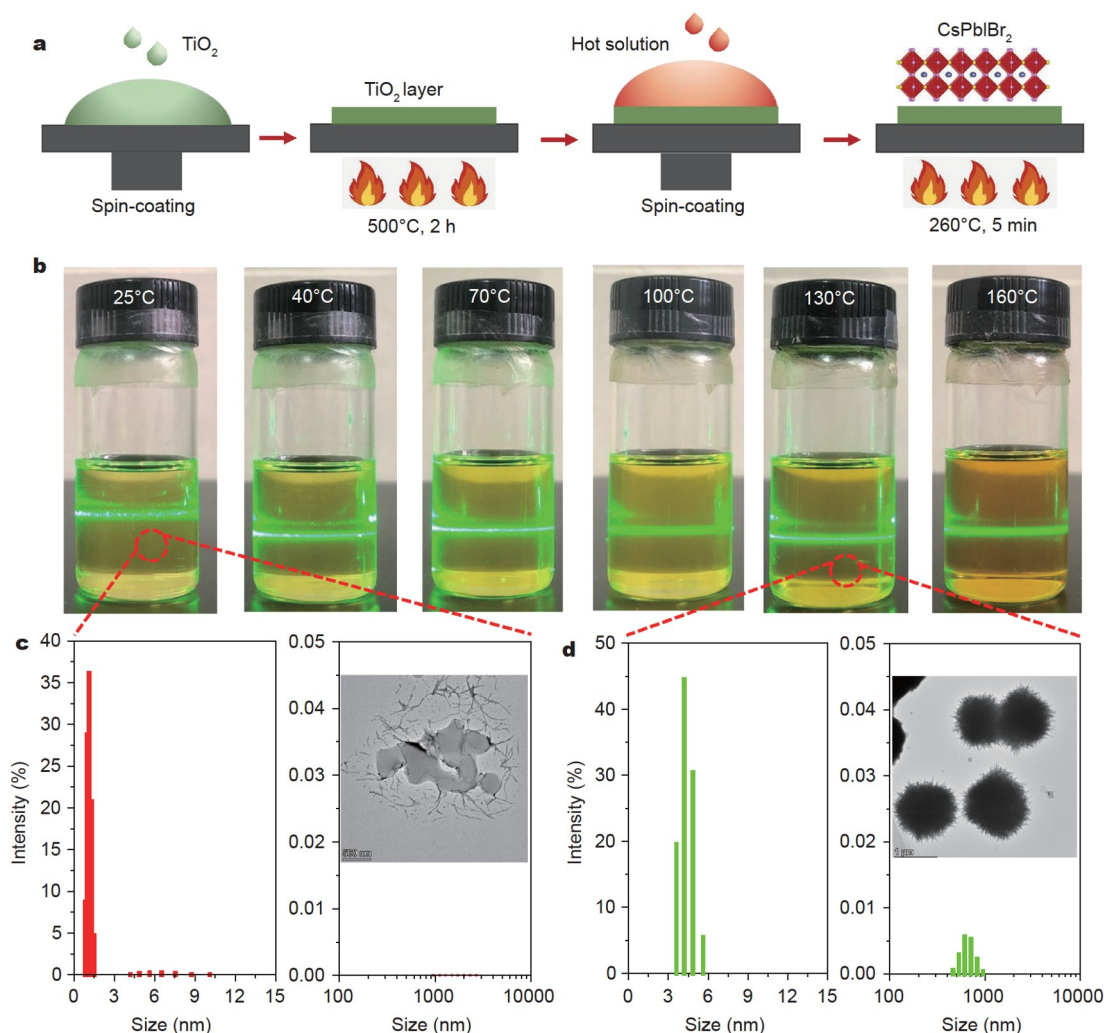


Figure 1 (a) CsPbI₂Br₂ film preparation. (b) Tyndall effect of perovskite precursor solutions at different temperatures. DLS results of perovskite precursors at (c) room temperature and (d) 130°C. The insets are TEM images of perovskite precursor solutions.

hot-solution-tailored film delivers a rapid phase conversion with an absorption wavelength of 550 nm, which is higher than 540 nm for the former case, ensuring the reduced energy barrier for crystal growth. As shown in Fig. S6, top-view and cross-sectional SEM images of perovskite films display the dependence of film quality and thickness on the temperature of the precursor solution. By plotting the evolution of grain size and film thickness as a function of temperature (Fig. 2c, d), the grain size gradually increases from an average of 340 to 850 nm as the temperature increases from 25 to 130°C, and further increasing the temperature to 160°C will reduce the grain size to 450 nm. In addition, higher temperature leads to a thicker film ranging from 165 to 280 nm because of the fast evaporation rate [25]. Finally, an optimized morphology for the control and microcrystal-induced perovskite films is provided in Fig. 2e, f. Meanwhile, based on the atomic force microscopic images shown in Fig. S7, the root-mean-squared value of the film (18.0 nm) tailored at 130°C is the lowest compared with the control and other ones. Therefore, the presence of microcrystals will promote the nucleation and growth of perovskite crystals.

In general, the formation of perovskite film is determined by the first nucleation step in a solution induced by the escape of

residual solvents and the subsequent downward crystal growth process [26,27]. As shown in Fig. 2g, low temperature suppresses the growth process, which leads to more nuclei. On the contrary, the high temperature accelerates the solvent evaporation and then increases the nucleus number. Therefore, suppressing the formation of extremely dense nuclei and increasing the crystal grain growth rate are important for high-quality perovskite film. The Gibbs free energy change (ΔG) of the system, along with the nuclei radius (r), is described by the volumetric (ΔG_V) and interfacial (ΔG_S) energies based on the following equation to comprehensively understand the kinetics of the crystal growth [28]:

$$G(r) = G_V + G_S = \left(-\frac{4\pi r^3}{3V_M} \right) RT \ln \left(\frac{c}{c_0} \right) + 4\pi r^2 \gamma,$$

where V_M is the volume of the nucleus, R and T are the gas constant and absolute temperature, respectively, and γ is the nucleus surface tension. As shown in Fig. 2h, large microcrystals are necessary to overcome the energy barrier and form nuclei. Compared with the control precursor, the presence of large microcrystals will change the nucleation mode from homogeneous (ΔG_{Hom}) to heterogeneous (ΔG_{Het}) kinetics. Conse-

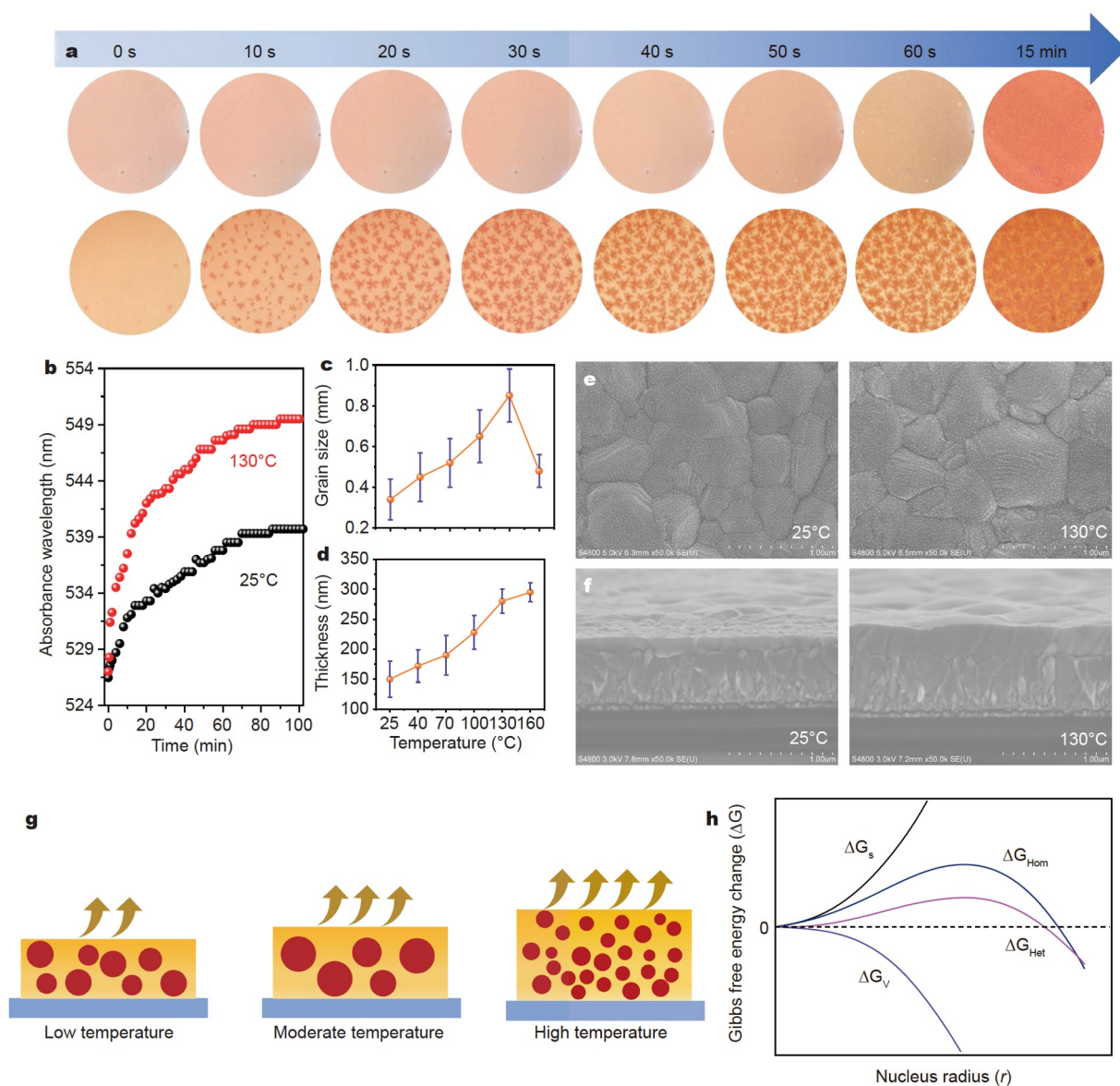


Figure 2 (a) Optical microscopic images of intermediate films at various times. (b) The evolution of characteristic absorption wavelengths of precursor films along with the standing time. (c) Grain sizes and (d) thicknesses of perovskite films fabricated at various temperatures. (e) Top-view and (f) cross-sectional SEM images of the control and optimal perovskite films. (g) Schematic illustration of the film growth at different temperatures. (h) The Gibbs energy change (ΔG) as a function of nucleus radius (r).

quently, the energy barrier is remarkably reduced, improving the crystal grain growth at moderate temperatures, which is consistent with the UV-vis absorption spectra (Fig. S8a) and XRD patterns (Fig. S8b). In detail, under thermal annealing, the residual solvent from the top surface of the “wet” film is evaporated to initialize the crystallization and growth of the intermediate phase on the film surface. Assisted by large high-temperature-induced microcrystals, grain growth is preferred from top to bottom, enlarging the grain size for the final perovskite film [27].

Based on the abovementioned characterization, we further fabricated the carbon-electrode-based CsPbIBr₂ PSC with the structure illustrated in Fig. 3a to study the photovoltaic performance evolution under various solution temperatures. Fig. 3b shows the typical J - V curves under one standard sun irradiation,

and the corresponding parameters, including open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), fill factor (FF), and PCE, are summarized in Table S1. With the increase in temperature, a significantly enhanced PCE up to 11.12% with V_{oc} of 1.323 V, J_{sc} of 11.53 mA cm⁻², and FF of 72.9% for a device tailored at 130°C is achieved, which is higher than that of the control device with 8.51% efficiency and consistent with the perovskite film quality. This is one of the highest efficiencies for all-inorganic CsPbIBr₂-based PSCs reported to date (Fig. 3c and Table S2) [29–42]. Furthermore, the distribution of statistical parameters from 20 devices for each case in Fig. 3d and Fig. S9 demonstrates excellent reproducibility. In addition, as shown in Fig. 3e, the integrated J_{sc} values deduced from the EQE spectra further demonstrate the high solar-to-electric conversion ability upon the introduction of large microcrystals into the precursor

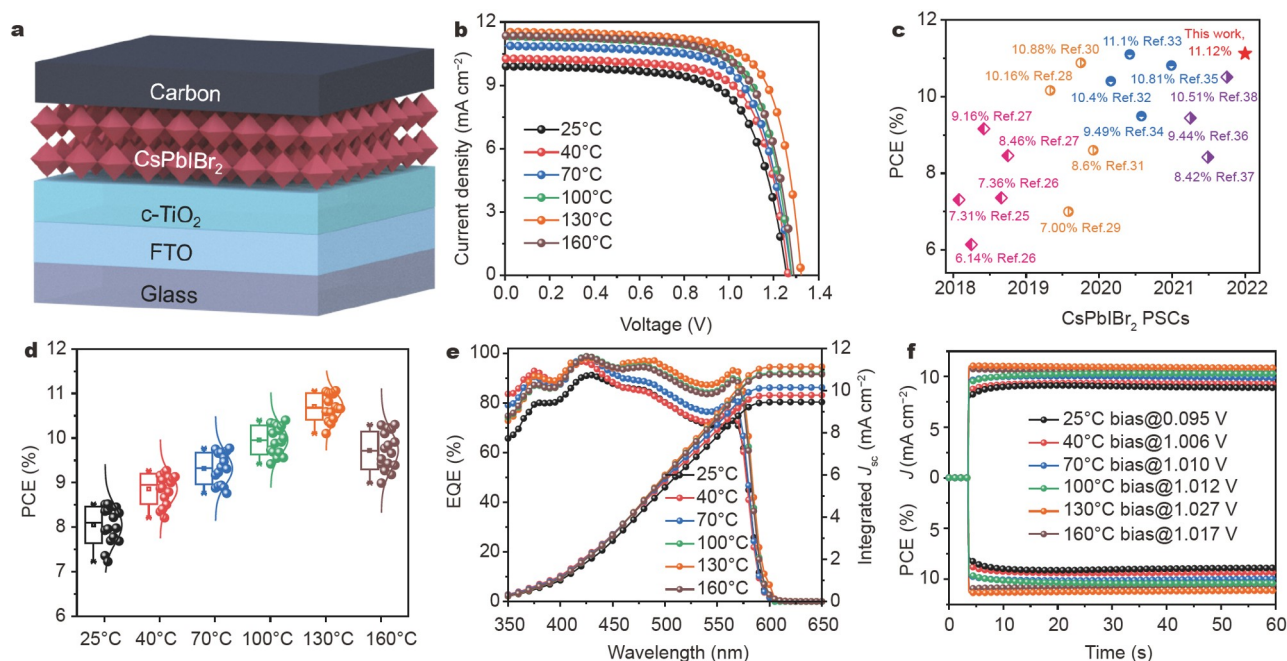


Figure 3 (a) Schematic structure of the PSC based on CsPbI₂Br₂ films. (b) Characteristic *J-V* curves of various PSCs. (c) Comparison of PCE with previously reported results. (d) Statistical PCE distribution, (e) EQE spectra, and (f) steady power output curves of various PSCs fabricated under various temperatures.

solution, which is in accordance with the results in *J-V* curves. In the presence of hysteresis (Fig. S10), the steady power outputs of various PSCs are recorded and shown in Fig. 3f, in which the steady PCEs of the control and hot-solution-tailored PSCs are consistent with the values obtained from *J-V* curves, indicating the reliable performance by regulating the precursor solution temperature. We have further characterized the *J-V* curves of PSCs based on the filtered solution under room and high temperatures to cross-check the role of microcrystals. As shown in Fig. S11, the efficiency of PSC is nearly unchanged after the filtration of the room-temperature solution. By contrast, the performance is significantly reduced for PSC in high-temperature cases, confirming the important role of large colloids in improving the photovoltaic performance of PSC.

The photovoltaic performance is highly dependent on the detrimental defects formed during the phase conversion and crystal growth. Space-charge-limited current measurements with electron-only structures were performed to quantize the defect density (N_t ; Fig. 4a). The smaller trap-filled limit voltage (V_{TFL}) suggests the reduced N_t from $5.68 \times 10^{16} \text{ cm}^{-3}$ for the film fabricated at 25°C to $1.29 \times 10^{16} \text{ cm}^{-3}$ for the film fabricated at 130°C based on the following relationship [43]: $N_t = 2\epsilon\epsilon_0 V_{TFL}/qL^2$, where q represents the elementary charge, L and ϵ are the thickness and relative dielectric constant of the CsPbI₂Br₂ film, and ϵ_0 is the vacuum permittivity. Consequently, the electron mobility is improved (Fig. S12) to deliver a higher FF. Then, the carrier transfer-recombination behavior was studied to reveal the intrinsic mechanism behind the performance enhancement. In general, for individual perovskite film under light irradiation, the photogenerated electrons will couple to radiate PL. Hence, the increased PL intensity of the microcrystal-induced perovskite film (Fig. S13) demonstrates that more electrons are radiatively recombined, cross-checking the reduced defects. The TRPL spectra shown in Fig. 4b are fitted on the basis of the

following equation [44]: $I = Ae^{-(t-t_0)/\tau_1} + Be^{-(t-t_0)/\tau_2}$, where I is the PL intensity, A and B are constants, τ_1 refers to the faster component of trap-mediated nonradiative recombination, and τ_2 is the slower component correlated to radiative recombination. The photogenerated carrier lifetimes (Table S3) are significantly increased for the perovskite films by elevating the solution temperature, which is consistent with the improved film quality. On the contrary, by transferring the perovskite film onto the TiO₂ surface, the evolution of the corresponding PL signal is reversed. As shown in Fig. 4c, d, we have recorded lifetime mapping images of the control and optimal samples. Evidently, a large dark region is observed among grains, which is primarily attributed to the rapid luminescence quenching when considering the substantial defects at the grain surface [45]. Compared with the control sample, the enlarged grain for the microcrystal-induced perovskite film improves the reduction of the detrimental recombination area and unwanted charge loss. In addition, the reduced PL intensity extracted from two perovskite grains (Fig. 4e, f) for the film tailored at 130°C demonstrates the rapid charge extraction rather than radiative recombination under the presence of the TiO₂ layer [46,47]. Therefore, the photovoltaic performance of PSC can be maximized when simultaneously considering the reduced defect and accelerated charge transfer.

Subsequently, we studied the recombination kinetics within PSC. As shown in Fig. 4g, the hot-solution-tailored device exhibits a higher built-in electric field (V_{bi}) than that of the control device by recording the capacitance-voltage (*C-V*) relationship based on the Mott-Schottky equation [48]: $C^{-2} = 2(V_{bi} - V)/(A^2 q_0 N_A)$, which undoubtedly delivers better diode characteristic for PSC. By plotting the dependence of V_{oc} and J_{sc} on light intensity (Fig. 4h and Fig. S14), the deviation of the slope from unity kT/q (where k is the Boltzmann constant) for the modified PSC is reduced, and α extracted from $J_{sc} \propto I^\alpha$ is closer

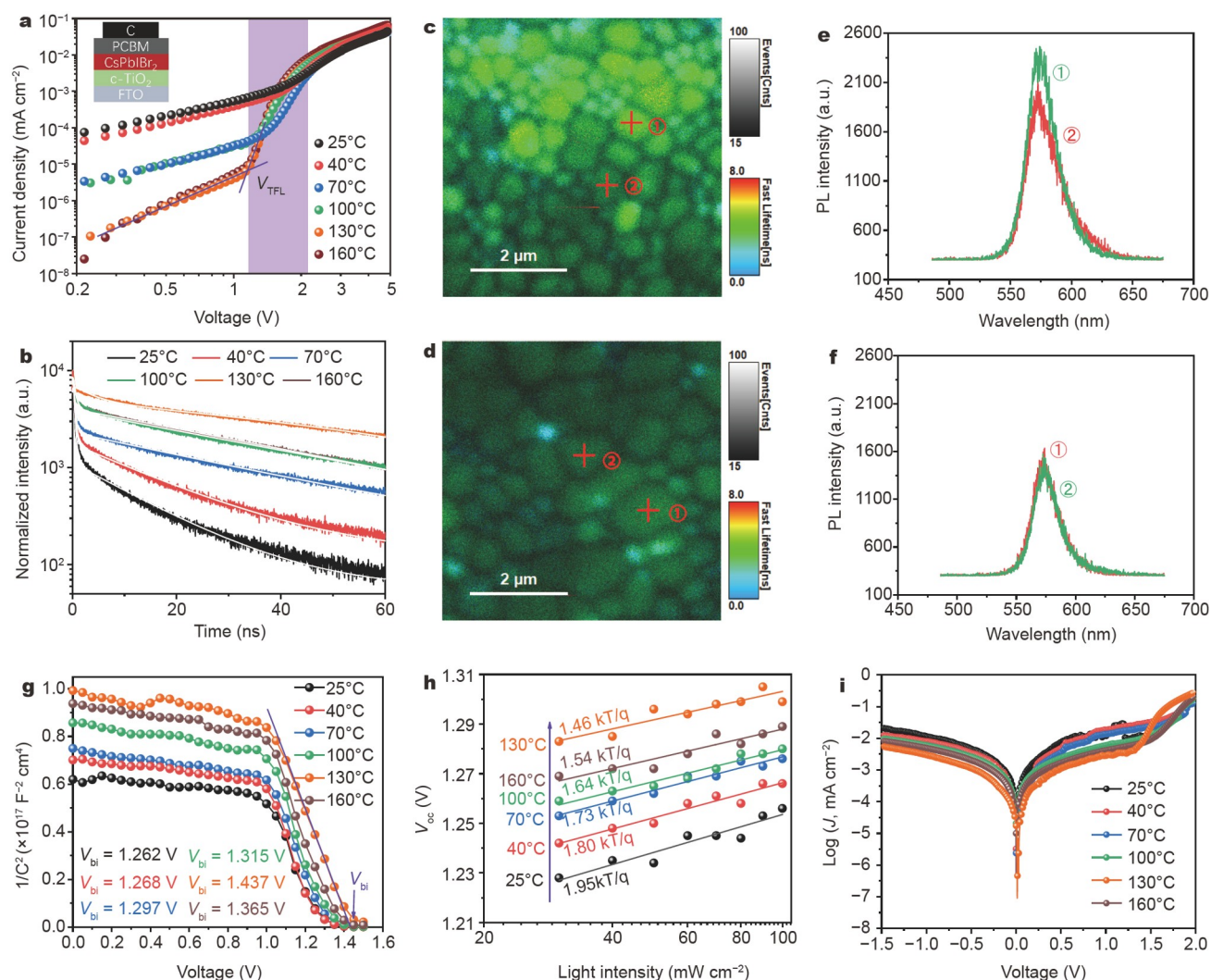


Figure 4 (a) Dark J - V curves of electron-only devices with a structure of FTO/ c -TiO₂/CsPbIBr₂/PCBM/carbon. (b) TRPL spectra of perovskite films on the TiO₂ surface. Lifetime mapping images of the (c) control and (d) optimal perovskite films on the TiO₂ surface and (e, f) PL spectra extracted from the corresponding points. (g) Mott-Schottky plots, (h) the dependence of V_{oc} on the light intensity, and (i) dark J - V curves of various PSCs.

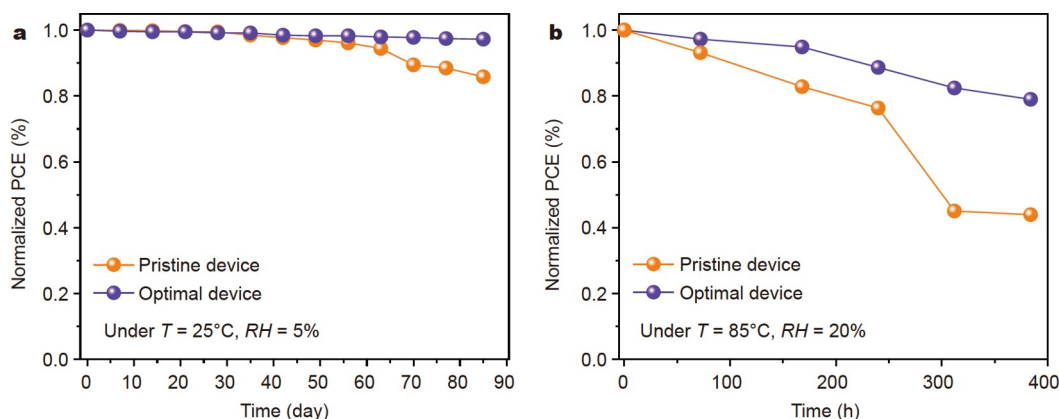


Figure 5 Long-term stability of pristine and best devices under (a) 25°C, 5% RH and (b) 85°C, 20% RH conditions.

to 1, indicating the greatly suppressed trap-assisted and bimolecular radiative recombination [49,50], which can also be cross-checked by the increased recombination resistance (Fig. S15) and prolonged electron lifetime within PSC (Fig. S16). Conse-

quently, the shunt current loss under dark conditions is significantly reduced in one order of magnitude (Fig. 4i), allowing for the maximal charge carrier scavenging.

Finally, the durability of PSC is evaluated by monitoring the

PCE under dry air with relative humidity (RH) of 5% and a high temperature of 85°C without encapsulation. As shown in Fig. 5a and Fig. S17, the PCE of the optimal device is nearly unchanged after 90 days of storage, whereas the control device can only maintain 80% of the initial performance, demonstrating improved stability. Compared with the traditional hybrid perovskite species, all-inorganic perovskites are highlighted by their excellent thermal stability because of the removal of unstable organic components. As shown in Fig. 5b and Fig. S18, a significantly improved stability under RH = 25%, $T = 85^\circ\text{C}$ for the optimal PSC is achieved over 400 h. Considering that the performance degradation is always triggered by detrimental defects, the high-quality perovskite film induced by solution microcrystal is undoubtedly responsible for this strengthened ability to withstand harsh stimuli [51].

CONCLUSIONS

We have reported an effective strategy to enhance the efficiency of all-inorganic CsPbI₂Br₂ PSC by elevating the precursor temperature to form a stable microcrystal, which will lower the energy barrier of heterogeneous nucleation and increase the growth rate of crystal grains. After comprehensive characterization, the resultant perovskite film with enlarged grains and reduced defects is obtained, which could improve the carrier transfer across the film to be collected by an underlying charge-selective layer. Finally, the intragranular charge recombination is suppressed, promoting a champion efficiency as high as 11.12% for a carbon-based device with significantly improved stability under ambient environment and high temperature. This work presents a distinct perspective and in-depth understanding of the properties of precursor solutions for regulating perovskite film quality, which can inspire us to develop highly efficient and stable PSCs and other advanced perovskite-based optoelectronic devices.

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Conflict of interest The authors declare that they have no conflict of interest.

Supplementary information Supporting data are available in the online version of the paper.



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溶液微晶提升全无机CsPbIBr₂钙钛矿的薄膜质量并减小碳基光伏器件内部载流子复合

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摘要 钙钛矿前驱体溶液的胶体特性决定了钙钛矿薄膜的质量以及钙钛矿太阳能电池的光电转换效率. 在本文中, 我们通过简单地调控钙钛矿溶液的温度, 实现了钙钛矿薄膜质量的优化. 系统的表征测试表明, 与室温下的钙钛矿前驱体溶液不同, 在高温条件下, 钙钛矿前驱体溶液中会产生尺寸在一微米以上的稳定微晶, 能够促进钙钛矿中间相的自发成核并实现薄膜的择优生长. 薄膜质量优化后, 由于缺陷态减少和晶粒尺寸增加, 光生电荷能够快速地在晶粒内部传输, 降低晶粒内部的非辐射复合以及辐射复合, 将电池效率提升至11.12%, 明显高于未优化器件的8.51%. 更重要的是, 优化器件表现出非常优异的空气稳定性和高温稳定性.